FIREWORK CHRONOS · DOCUMENT 2 OF 2

APIs, Keys and Datasets

Every formula input, its free source and backups, the keys to create, what each API returns field by field, and what changes with time.



Dhruv Sharma and team · 20 October 2026

Free tiers change often — each limit carries the date it was checked.

Contents

1 The bottom line **4**

1.1 What this document gives you 4

1.2 Formula inputs at a glance 4

2 Keys and accounts to create **6**

2.1 The key list 6

3 Every input x, its sources and backups **8**

3.1 `x_fire_points` — Active-fire detections (MODIS + VIIRS) 8

3.2 `x_clear_days` — Days each cell was visible (not cloud) - daily FireMask 8

3.3 `x_burned_area` — Burned area per month (blind-spot check) 9

3.4 `x_land_cover` — Cropland / tree / built-up class around a fire 9

3.5 `x_industry` — Industrial sites, kilns, flares near a fire 10

3.6 `x_weather` — Wind, humidity, rain, temperature for context 10

3.7 `x_air_quality` — PM2.5 near the burning (impact panel) 11

3.8 `x_orbits` — Satellite orbit elements -> overpass times 11

3.9 `x_boundaries` — District / state boundaries for regions and briefs 12

3.10 `x_ndvi` — Vegetation greenness / harvest timing 12

3.11 `x_granule_index` — Which product files exist for a date (health + archive) 13

4 What each API returns **14**

4.1 FIRMS Area API — VIIRS (CSV) 14

4.2 FIRMS Area API — MODIS (CSV) 14

4.3 FIRMS MAP_KEY status 15

4.4 FIRMS data availability 15

4.5 NASA CMR granule search 15

4.6 Earth Engine — monthly clear and fire days 16

4.7 Earth Engine — FireMask classes 17

4.8 Earth Engine — burned area (MCD64A1) 17

4.9 Earth Engine — land cover (WorldCover) 18

4.10 Open-Meteo forecast 18

4.11 Open-Meteo archive 19

4.12 Open-Meteo air quality 20

4.13 NASA POWER daily point 20

4.14 OpenAQ — nearby stations 21

4.15 OpenAQ — measurements 22

4.16 Overpass (OpenStreetMap) 22

4.17 CelesTrak orbit elements 23
4.18 geoBoundaries (REAL response) 24
4.19 LLM chat (OpenAI-compatible) 24
4.20 Gemini generateContent 25
4.21 Groq Whisper transcription 25
4.22 Embeddings (OpenAI-compatible) 26
4.23 Our own API (the contract) 26

5 What changes with time and what does not 28

6 When a source fails 29
6.1 Data kit files 29

OVERVIEW

1 The bottom line

Which data each formula needs, where to get it, and what to do when a source fails.

1.1 What this document gives you

Every formula in the FIREWORK atlas needs some inputs — call each one an **x**. The fire record needs fire points; the clear-sky rate needs the number of days a place was visible; the blind-spot flag needs burned area; source attribution needs land cover and industry maps; the overpass clock needs orbit data. This document lists those eleven inputs, the free API or dataset for each, at least one backup, the key you need, the limits, how often the data changes, and exactly what the response looks like when you call the API.

The same list is also saved as a machine-readable file, `data_sources.json`, in the data kit. The small resolver `fx_engine.py` reads that file: for an input x it tries the first source, checks that the answer is healthy and complete, and if not, moves to the next source automatically. Every answer comes back as `{data, meta}`, where `meta.data_source` tells you which source was really used. That is how the website stays honest even when an API is down.

Bottom line

- **Must-have keys (all free):** NASA Earthdata login, FIRMS MAP_KEY, Google Earth Engine non-commercial project, OpenAQ key.
- **No key needed:** Open-Meteo, NASA POWER, CelesTrak, Overpass (OSM), geoBoundaries, NASA CMR search, Planetary Computer (read).
- **Biggest data:** fire points (FIRMS archive) and daily fire masks (Earth Engine). Download once, aggregate to cell-days, store as Parquet — never call the API for history on every page load.
- **Every input has a backup;** most have two.

1.2 Formula inputs at a glance

INPUT X	MEANING	FORMULAS	PRIMARY	BACKUPS
`x_fire_points`	Active-fire detections (MODIS + VIIRS)	F01, F02, F03, F05, F06, F07, F08, F13, F20, F21, F44	FIRMS Area API	FIRMS archive download (Download Request); LAADS DAAC / earthaccess (MCD14ML, VNP14IMG, VJ114IMG)
`x_clear_days`	Days each cell was visible (not cloud) - daily FireMask	F09, F25	Google Earth Engine MOD14A1 / MYD14A1 / VNP14A1	Microsoft Planetary Computer modis-14A1-061; NASA AppEEARS area request
`x_burned_area`	Burned area per month (blind-spot check)	F46	GEE MODIS/061/MCD64A1	ORNL DAAC MODIS subsets (MCD64A1); ESA FireCCI

INPUT X	MEANING	FORMULAS	PRIMARY	BACKUPS
x_land_cover	Cropland / tree / built-up class around a fire	F08, F42, F43	ESA WorldCover 10 m (GEE)	ESA WorldCover on AWS S3; MODIS MCD12Q1 yearly land cover
x_industry	Industrial sites, kilns, flares near a fire	F08, F42, F43	OSM Overpass API	Overpass mirror (private.coffee / maps.mail.ru); Geofabrik India extract (.osm.pbf)
x_weather	Wind, humidity, rain, temperature for context	context panels	Open-Meteo Forecast + Archive	NASA POWER daily point; Copernicus CDS ERA5
x_air_quality	PM2.5 near the burning (impact panel)	impact panel	OpenAQ API v3	Open-Meteo Air Quality (CAMS model); CPCB / WAQI
x_orbits	Satellite orbit elements → overpass times	F01, F45	CelesTrak GP API	Space-Track; Derive from FIRMS acq_time
x_boundaries	District / state boundaries for regions and briefs	F04 (areas), region totals	geoBoundaries API	Natural Earth; GADM
x_ndvi	Vegetation greenness / harvest timing	season context (F29)	GEE MODIS/061/MOD13Q1	ORNL MODIS subsets; Sentinel-2 via Earth Search STAC
x_granule_index	Which product files exist for a date (health + archive)	pipeline / health page	NASA CMR Search API	earthaccess Python library

ACCOUNTS

2 Keys and accounts to create

All free. Create them early — some take days to approve.

2.1 The key list

Each team member should create their own accounts. Rate limits are counted per key or per project, so sharing one key means one person's test run can block everyone else during the hackathon. Store keys only in environment variables (a `.env` file that is listed in `.gitignore`, and the hosting platform's secret settings). Never paste a key into frontend code: anything shipped to the browser can be read by anyone.

SERVICE	SIGN-UP	UNLOCKS	FREE LIMIT	APPROVAL
NASA Earthdata Login	`https://urs.earthdata.nasa.gov/users/new`	FIRMS archive download, LAADS, AppEEARS, earthaccess	free; needed for bulk files	instant
FIRMS MAP_KEY	`https://firms.modaps.eosdis.nasa.gov/api/map_key/`	FIRMS Area API (fires)	5,000 transactions / 10 min; 1–5 days per call	instant (email)
Google Earth Engine (noncommercial)	`https://earthengine.google.com/noncommercial/`	Fire masks, burned area, land cover, NDVI	Community tier 150 EECU-hours/month; Contributor 1,000 (needs billing account, no charge); quotas since 27 Apr 2026	hours–days
OpenAQ v3	`https://explore.openaq.org/register`	PM2.5 stations	60 requests/min, 2,000/hour	instant
Copernicus CDS	`https://cds.climate.copernicus.eu`	ERA5 weather (backup)	queued jobs	instant
Space-Track	`https://www.space-track.org/auth/createAccount`	Orbit elements (backup)	account limits	1–2 days
EOG (Colorado School of Mines)	`https://eogdata.mines.edu/products/vnf/`	VIIRS Nightfire / gas flares	file downloads	1–2 days
WAQI token	`https://aqicn.org/data-platform/token/`	Air quality backup	token quota	instant

⚠ Key safety rules

1. Keys live in `.env` (local) and platform secrets (deployed). `.env` is in `.gitignore`.
2. The browser never sees a key: the frontend calls our backend, the backend calls NASA.
3. One key per person; rotate a key immediately if it was pushed to GitHub.
4. Log which key and which source answered each request (meta) so problems can be traced.

Key = ghar ki chaabi. Browser mein kabhi mat rakho — sirf server ke paas.

INPUTS

3 Every input x, its sources and backups

Primary first, then backups in the order the resolver tries them.

3.1 x_fire_points — Active-fire detections (MODIS + VIIRS)

Fire points are the core of the project: every hot pixel MODIS and VIIRS reported. For the last few days use the FIRMS Area API; for history (2000 to today) do one bulk download from the FIRMS archive and keep it in your own storage. Near-real-time (NRT) rows are replaced by standard-processing (SP) rows a few months later, so history must be refreshed from the archive, not the NRT feed.

Used by: F01, F02, F03, F05, F06, F07, F08, F13, F20, F21, F44 **Unit:** rows (one per hot pixel)

Changes over time: NRT rows appear within ~3 h; Standard Processing (SP) replaces NRT after ~2-3 months; archive versions (MODIS C6.1, VIIRS 002) can be reprocessed

#	SOURCE	KEY	LIMIT	BEST FOR
1	FIRMS Area API https://firms.modaps.eosdis.nasa.gov/api/area/	FIRMS_MAP_KEY	5000 transactions / 10 min, 1-5 days per call	recent days, small areas
2	FIRMS archive download (Download Request) https://firms.modaps.eosdis.nasa.gov/download/	Earthdata login	emailed CSV/SHP, years at once	2000→today history
3	LAADS DAAC / earthaccess (MCD14ML, VNP14IMG, VJ114IMG) https://ladsweb.modaps.eosdis.nasa.gov/	Earthdata login	bulk files	raw product files when FIRMS is down
4	GEE community catalog FIRMS vectors https://gee-community-catalog.org/projects/firms_vector/	Earth Engine project	EE quota	server-side aggregation

3.2 x_clear_days — Days each cell was visible (not cloud) - daily FireMask

To turn fire counts into a fair rate we need to know on how many days each cell could be seen. The daily fire-mask products classify every pixel as cloud, clear land, water or fire. Earth Engine can count clear and fire days per cell per month on Google's servers and return only a small table — that is the optimisation that makes this feasible for free.

Used by: F09, F25 **Unit:** clear cell-days

Changes over time: daily product; fixed after processing; new collection versions rare

#	SOURCE	KEY	LIMIT	BEST FOR
1	Google Earth Engine MOD14A1 / MYD14A1 / VNP14A1 https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MOD14A1	Earth Engine project	Community tier 150 EECU-h/month	monthly clear-day counts per cell
2	Microsoft Planetary Computer modis-14A1-061 https://planetarycomputer.microsoft.com/dataset/modis-14A1-061	none for read	fair use	backup, STAC + COG
3	NASA AppEEARS area request https://appeears.earthdatacloud.nasa.gov/	Earthdata login	queued jobs	small-area subsets

3.3 x_burned_area — Burned area per month (blind-spot check)

Burned area is an independent measure of burning that does not depend on catching the fire in the act. It is the input for the blind-spot ratio. It arrives monthly and a few months late, so the newest months show 'pending' instead of zero.

Used by: F46 **Unit:** km2 or burned pixels

Changes over time: monthly; available ~2-3 months after the month ends

#	SOURCE	KEY	LIMIT	BEST FOR
1	GEE MODIS/061/MCD64A1 https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MCD64A1	Earth Engine project	EE quota	monthly burned km2 per district
2	ORNL DAAC MODIS subsets (MCD64A1) https://modis.ornl.gov/	none	point/small box	backup for single sites
3	ESA FireCCI https://climate.esa.int/en/projects/fire/	none	file download	independent cross-check

3.4 x_land_cover — Cropland / tree / built-up class around a fire

Land cover tells whether a fire sits in cropland, forest or a built-up area. It changes slowly, so it is downloaded once and stored, not requested live.

Used by: F08, F42, F43 **Unit:** class code

Changes over time: static snapshot (WorldCover 2020, 2021); MCD12Q1 yearly

#	SOURCE	KEY	LIMIT	BEST FOR
1	ESA WorldCover 10 m (GEE) https://developers.google.com/earth-engine/datasets/catalog/ESA_WorldCover_v200	Earth Engine project	EE quota	cropland share per cell
2	ESA WorldCover on AWS S3 https://registry.opendata.aws/esa-worldcover-vito/	none	free download	offline backup

#	SOURCE	KEY	LIMIT	BEST FOR
3	MODIS MCD12Q1 yearly land cover https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MCD12Q1	Earth Engine or Earthdata	EE quota	years before 2020

3.5 x_industry — Industrial sites, kilns, flares near a fire

Industrial sites, brick kilns and gas flares cause hot pixels all year. OpenStreetMap has many of them; the gas-flare list adds the rest. Query once per region and store the result.

Used by: F08, F42, F43 **Unit:** points/polygons

Changes over time: OSM edited daily by volunteers; flare lists yearly

#	SOURCE	KEY	LIMIT	BEST FOR
1	OSM Overpass API https://overpass-api.de/api/interpreter	none	fair use, ~2 slots per IP	industrial landuse, kilns
2	Overpass mirror (private.coffee / maps.mail.ru) https://wiki.openstreetmap.org/wiki/Overpass_API#Public_Overpass_API_instances	none	fair use	when main server is busy
3	Geofabrik India extract (.osm.pbf) https://download.geofabrik.de/asia/india.html	none	daily file	offline, no rate limit
4	VIIRS Nightfire / EOG gas flares https://eogdata.mines.edu/products/vnf/	free EOG account	file download	flare masks

3.6 x_weather — Wind, humidity, rain, temperature for context

Weather explains why a season was busy or quiet: dry, windy days spread fire; rain stops it. Weather is context, not part of the core calculation, so a free no-key source is ideal.

Used by: context panels **Unit:** hourly/daily values

Changes over time: forecast changes every hour; archive (ERA5) final after ~5 days

#	SOURCE	KEY	LIMIT	BEST FOR
1	Open-Meteo Forecast + Archive https://open-meteo.com/en/docs	none (non-commercial)	about 10,000 calls/day non-commercial	fast JSON, no key
2	NASA POWER daily point https://power.larc.nasa.gov/docs/services/api/	none	fair use	long daily history
3	Copernicus CDS ERA5 https://cds.climate.copernicus.eu	CDS API key	queued jobs	research-grade grids

3.7 x_air_quality — PM2.5 near the burning (impact panel)

Air quality connects fires to people. Ground stations are best where they exist; model data fills the gaps but must be labelled as modelled.

Used by: impact panel **Unit:** ug/m3

Changes over time: hourly; stations go offline; values can be corrected later

#	SOURCE	KEY	LIMIT	BEST FOR
1	OpenAQ API v3 https://docs.openaq.org	X-API-Key (free)	60/min, 2000/hour	ground stations
2	Open-Meteo Air Quality (CAMS model) https://open-meteo.com/en/docs/air-quality-api	none	non-commercial fair use	model values everywhere
3	CPCB / WAQI https://aqicn.org/data-platform/token/	WAQI token	token quota	India backup

3.8 x_orbits — Satellite orbit elements -> overpass times

Orbit elements let us compute when each satellite passes over a place — needed for the overpass clock and the diurnal blind-spot formula. Elements age quickly; refresh daily and cache.

Used by: F01, F45 **Unit:** OMM/TLE

Changes over time: updated several times a day; old elements drift - refresh daily

#	SOURCE	KEY	LIMIT	BEST FOR
1	CelesTrak GP API https://celestrak.org/NORAD/documentation/gp-data-formats.php	none	fair use; cache 2 h+	JSON elements
2	Space-Track https://www.space-track.org	free account	account limits	official backup
3	Derive from FIRMS acq_time -	none	-	offline estimate

3.9 x_boundaries — District / state boundaries for regions and briefs

Boundaries turn cells into districts and states for briefs and totals. They are almost static; download once and simplify for the web.

Used by: F04 (areas), region totals **Unit:** GeoJSON polygons

Changes over time: static; new districts appear a few times a year

#	SOURCE	KEY	LIMIT	BEST FOR
1	geoBoundaries API https://www.geoboundaries.org/api.html	none	fair use	ADM1/ADM2 GeoJSON
2	Natural Earth https://www.naturalearthdata.com	none	download	light world maps
3	GADM https://gadm.org	none (non-commercial)	download	detailed backup

3.10 x_ndvi — Vegetation greenness / harvest timing

Vegetation greenness drops sharply after harvest. It helps explain timing of the burning season. Optional context.

Used by: season context (F29) **Unit:** NDVI -1..1

Changes over time: 16-day composites

#	SOURCE	KEY	LIMIT	BEST FOR
1	GEE MODIS/061/MOD13Q1 https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MOD13Q1	Earth Engine project	EE quota	250 m NDVI
2	ORNL MODIS subsets https://modis.ornl.gov/	none	small areas	backup
3	Sentinel-2 via Earth Search STAC https://earth-search.aws.element84.com/v1	none	fair use	10 m detail

3.11 `x_granule_index` — Which product files exist for a date (health + archive)

The granule index tells which product files exist for a given day. It powers the health page and lets the pipeline know when new archive data has arrived.

Used by: pipeline / health page **Unit:** file metadata

Changes over time: new granules every orbit

#	SOURCE	KEY	LIMIT	BEST FOR
1	NASA CMR Search API https://cmr.earthdata.nasa.gov/search/site/docs/search/api.html	none for search	fair use	list files
2	earthaccess Python library https://earthaccess.readthedocs.io	Earthdata login	-	search + download in Python

RESPONSES

4 What each API returns

Launch the call, get this. Field by field.

This chapter shows, for each API, the exact call and what comes back. The response shapes follow each provider's official documentation. The numbers inside are illustrative — they show the format, not real measurements — except the geoBoundaries response, which is a real response fetched on 20 October 2026. The data kit contains `fetch_samples.py`: run it on your laptop with your keys and it saves the real responses into `real_samples/`, so you can compare them with these pages.

4.1 FIRMS Area API — VIIRS (CSV)

Call: `GET /api/area/csv/{MAP_KEY}/VIIRS_NOAA20_NRT/73,29,77,32/2`

Key: FIRMS MAP_KEY **Freshness:** Every 3 h (NRT); SP replaces later **File:** `samples/firms_viirs_area.csv`

```
latitude,longitude,bright_ti4,scan,track,acq_date,acq_time,satellite,instrument,confidenc...
30.91234,75.84567,341.25,0.45,0.39,2024-11-12,0806,N20,VIIRS,n,2.0NRT,296.10,7.85,D
30.88021,75.90112,355.62,0.45,0.39,2024-11-12,0806,N20,VIIRS,h,2.0NRT,298.44,15.20,D
```

FIELD	MEANING / HOW WE USE IT
`latitude, longitude`	pixel centre (not the fire outline)
`bright_ti4 / bright_ti5`	brightness temperature in kelvin, I4 (3.7 μm) and I5 (11 μm)
`scan, track`	pixel size in km across and along track (F02)
`acq_date, acq_time`	UTC date and HHMM time (convert with F01)
`satellite`	N = Suomi NPP, N20 = NOAA-20, N21 = NOAA-21
`confidence`	l / n / h (F07)
`version`	2.0NRT = near real time; 2 = standard processing
`frp`	fire radiative power, MW (F21)
`daynight`	D or N (F13, F44)

4.2 FIRMS Area API — MODIS (CSV)

Call: `GET /api/area/csv/{MAP_KEY}/MODIS_NRT/73,29,77,32/2`

Key: FIRMS MAP_KEY **Freshness:** Every 3 h; SP later **File:** `samples/firms_modis_area.csv`

```
latitude,longitude,brightness,scan,track,acq_date,acq_time,satellite,instrument,confidenc...
30.9051,75.8712,318.4,1.1,1.0,2024-11-12,0812,Aqua,MODIS,72,6.1NRT,297.2,21.6,D
```

FIELD	MEANING / HOW WE USE IT
`brightness / bright_t31`	kelvin, channel 21/22 and channel 31
`confidence`	0–100 (keep $\geq$ 30, F07)
`satellite`	Terra or Aqua
`version`	6.1NRT or 6.1
`type (archive only)`	0 vegetation, 1 volcano, 2 static land source, 3 offshore

4.3 FIRMS MAP_KEY status

Call: `GET /mapserver/mapkey_status/?MAP_KEY=…`

Key: FIRMS MAP_KEY **Freshness:** live **File:** `samples/firms_mapkey_status.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_endpoint": "GET https://firms.modaps.eosdis.nasa.gov/mapserver/mapkey_status/?MAP_KEY=...
 "transaction_limit": 5000,
 "current_transactions": 12,
 "transaction_interval": "10 minutes"
}
```


FIELD	MEANING / HOW WE USE IT
`transaction_limit`	5,000 per interval
`current_transactions`	used in this window — slow down near the limit
`transaction_interval`	10 minutes

4.4 FIRMS data availability

Call: `GET /api/data_availability/csv/{MAP_KEY}/all`

Key: FIRMS MAP_KEY **Freshness:** daily **File:** `samples/firms_data_availability.csv`


```
data_id,min_date,max_date
MODIS_NRT,2026-08-01,2026-10-19
MODIS_SP,2000-11-01,2026-07-31
VIIRS_SNPP_NRT,2026-07-01,2026-10-19
VIIRS_SNPP_SP,2012-01-20,2026-06-30
VIIRS_NOAA20_NRT,2026-07-01,2026-10-19
VIIRS_NOAA20_SP,2018-04-01,2026-06-30
VIIRS_NOAA21_NRT,2024-01-17,2026-10-19
```


FIELD	MEANING / HOW WE USE IT
`data_id`	source name to use in Area API calls
`min_date / max_date`	date range available — use it to switch from SP to NRT

4.5 NASA CMR granule search

Call: `GET cmr.earthdata.nasa.gov/search/granules.json?short_name=VNP14IMG&temporal=…&bounding_box=…`

Key: none **Freshness:** every orbit **File:** `samples/cmr_granules.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_endpoint": "GET https://cmr.earthdata.nasa.gov/search/granules.json?short_name=VNP14IM...
 "feed": {
  "updated": "2026-10-20T06:00:00.000Z",
  "id": "https://cmr.earthdata.nasa.gov/search/granules.json?...",
  "title": "ECHO granule metadata",
  "entry": [
   {
    "id": "G0000000000-LANCEMODIS",
    "title": "VNP14IMG.A2024317.0806.002.2024317120000.nc",
    "time_start": "2024-11-12T08:06:00.000Z",
    "time_end": "2024-11-12T08:12:00.000Z",
    "day_night_flag": "DAY",
    "granule_size": "1.2",
    "collection_concept_id": "C0000000000-LANCEMODIS",
    "online_access_flag": true,
    "links": [
     {
```

```
      "rel": "http://esipfed.org/ns/fedsearch/1.1/data#",
      "href": "https:// .../VNP14IMG.A2024317.0806.002.nc"
     }
    ]
   }
  ]
}
  ... (1 more lines in samples/cmr_granules.json)
```


FIELD	MEANING / HOW WE USE IT
`entry[].title`	file name; contains date (A2024317 = year + day of year) and time
`time_start / time_end`	pass time in UTC
`day_night_flag`	DAY / NIGHT
`links[].href`	download link (needs Earthdata login)

4.6 Earth Engine — monthly clear and fire days

Call: `ee.ImageCollection('MODIS/061/MYD14A1') … reduceRegions(cells)`

Key: EE project **Freshness:** daily images; we export monthly **File:** `samples/gee_firemask_month_reduce.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_code": "ee.ImageCollection('MODIS/061/MYD14A1').filterDate('2024-11-01','2024-12-01')....
 "type": "FeatureCollection",
 "features": [
  {
   "type": "Feature",
   "id": "1209_2558",
   "properties": {
    "cell": "1209_2558",
    "clear_days": 22,
    "fire_days": 14,
    "sensor": "MYD14A1",
    "month": "2024-11"
   }
  }
 ]
}
```


FIELD	MEANING / HOW WE USE IT
`clear_days`	days with FireMask 5, 7, 8 or 9 — denominator K of F09
`fire_days`	days with FireMask 7–9 — numerator F
`cell, month, sensor`	join keys for our database

4.7 Earth Engine — FireMask classes

Call: `band FireMask of MOD14A1 / MYD14A1 / VNP14A1`

Key: EE project **Freshness:** fixed definition **File:** `samples/gee_firemask_classes.json`


```
{
 "FireMask": {
  "0": "not processed (missing input)",
  "1": "not processed (obsolete)",
  "2": "not processed (other)",
  "3": "non-fire water",
  "4": "cloud",
  "5": "non-fire land",
  "6": "unknown",
  "7": "fire (low confidence)",
  "8": "fire (nominal confidence)",
  "9": "fire (high confidence)"
 },
 "MaxFRP": "MW x 10 (scale 0.1)",
 "_rule": "clear = class 5,7,8,9 ; fire = class 7,8,9 (or 8,9 for stricter)"
}
```


FIELD	MEANING / HOW WE USE IT
`4`	cloud — the day is not counted as observed
`5`	clear land, no fire
`7, 8, 9`	fire low / nominal / high confidence
`MaxFRP`	scale 0.1 → MW

4.8 Earth Engine — burned area (MCD64A1)

Call: `ee.ImageCollection('MODIS/061/MCD64A1') … reduceRegions(districts)`

Key: EE project **Freshness:** monthly, ~2–3 months late **File:** `samples/gee_mcd64a1_district.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_bands": {
  "BurnDate": "day of year burned (0 = unburned, -1 = unmapped, -2 = water)",
  "Uncertainty": "days",
  "QA": "bit flags",
  "FirstDay": "doy",
  "LastDay": "doy"
 },
 "type": "FeatureCollection",
 "features": [
  {
   "type": "Feature",
   "properties": {
    "district": "Ludhiana",
    "month": "2024-11",
    "burned_pixels_500m": 9132,
    "burned_km2": 2283.0
   }
  }
 ]
}
```


FIELD	MEANING / HOW WE USE IT
`BurnDate`	day of year of burning; 0 unburned, −1 unmapped, −2 water
`burned_km2`	pixels × 0.25 km^2 (500 m pixels) — input BA of F46

4.9 Earth Engine — land cover (WorldCover)

Call: `ee.ImageCollection('ESA/WorldCover/v200') … frequencyHistogram`

Key: EE project **Freshness:** static (2020, 2021) **File:** `samples/gee_worldcover_cell.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_classes": {
  "10": "tree cover",
  "20": "shrubland",
  "30": "grassland",
  "40": "cropland",
  "50": "built-up",
  "60": "bare / sparse",
  "70": "snow and ice",
  "80": "permanent water",
  "90": "herbaceous wetland",
  "95": "mangroves",
  "100": "moss and lichen"
 },
 "cell": "1209_2558",
 "histogram": {
  "40": 0.81,
  "50": 0.09,
  "10": 0.04,
  "80": 0.02,
  "30": 0.04
 }
}
```


FIELD	MEANING / HOW WE USE IT
`histogram`	share of each class in the cell; 40 = cropland, 50 = built-up, 10 = trees
`use`	agricultural exposure AE in F42

4.10 Open-Meteo forecast

Call: `GET api.open-meteo.com/v1/forecast?latitude=…&longitude=…&hourly=…`

Key: none **Freshness:** hourly model runs **File:** `samples/open_meteo_forecast.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_endpoint": "GET https://api.open-meteo.com/v1/forecast?latitude=30.9&longitude=75.85&h...
 "latitude": 30.875,
 "longitude": 75.875,
 "generationtime_ms": 0.05,
 "utc_offset_seconds": 0,
 "timezone": "GMT",
 "timezone_abbreviation": "GMT",
 "elevation": 247.0,
 "hourly_units": {
  "time": "iso8601",
  "temperature_2m": "degC",
  "relative_humidity_2m": "%",
  "wind_speed_10m": "km/h"
 },
 "hourly": {
  "time": [
   "2026-10-20T00:00",
   "2026-10-20T01:00",
   "2026-10-20T02:00"
  ],
  "temperature_2m": [
   19.4,
   18.9,
   18.5
  ... (13 more lines in samples/open_meteo_forecast.json)
```

FIELD	MEANING / HOW WE USE IT
`hourly.time[]`	ISO times (UTC unless timezone set)
`hourly.<variable>[]`	values aligned with time[]
`hourly_units`	unit of each variable
`latitude/longitude`	nearest grid point (may differ slightly from your input)

4.11 Open-Meteo archive

Call: `GET archive-api.open-meteo.com/v1/archive?…&daily=…`

Key: none **Freshness:** final after ~5 days **File:** `samples/open_meteo_archive.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_endpoint": "GET https://archive-api.open-meteo.com/v1/archive?latitude=30.9&longitude=...
 "latitude": 30.9,
 "longitude": 75.85,
 "timezone": "Asia/Kolkata",
 "elevation": 247.0,
 "daily_units": {
  "time": "iso8601",
  "temperature_2m_max": "degC",
  "precipitation_sum": "mm",
  "wind_speed_10m_max": "km/h"
 },
 "daily": {
  "time": [
   "2024-11-01",
   "2024-11-02"
  ],
  "temperature_2m_max": [
   29.8,
   29.1
  ],
  "precipitation_sum": [
   0.0,
   0.0
  ],
  ... (6 more lines in samples/open_meteo_archive.json)
```


FIELD	MEANING / HOW WE USE IT
`daily.time[]`	dates
`daily.<variable>[]`	daily max/sum/mean as requested

4.12 Open-Meteo air quality

Call: `GET air-quality-api.open-meteo.com/v1/air-quality?…`

Key: none **Freshness:** hourly (model) **File:** `samples/open_meteo_air_quality.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_endpoint": "GET https://air-quality-api.open-meteo.com/v1/air-quality?latitude=30.9&lo...
 "latitude": 30.9,
 "longitude": 75.9,
 "hourly_units": {
  "time": "iso8601",
  "pm2_5": "ug/m3",
  "pm10": "ug/m3",
  "carbon_monoxide": "ug/m3"
 },
 "hourly": {
  "time": [
   "2026-10-20T00:00",
   "2026-10-20T01:00",
   "2026-10-20T02:00"
  ],
  "pm2_5": [
   88.1,
   92.4,
   97.0
  ],
  "pm10": [
   140.2,
   146.8,
   151.3
  ... (8 more lines in samples/open_meteo_air_quality.json)
```


FIELD	MEANING / HOW WE USE IT
`pm2_5, pm10`	modelled concentration, μg/m^3 — label as model, not station

4.13 NASA POWER daily point

Call: `GET power.larc.nasa.gov/api/temporal/daily/point?parameters=T2M,RH2M,WS2M,PRECTOTCORR…`

Key: none **Freshness:** daily, a few days late **File:** `samples/nasa_power_daily.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_endpoint": "GET https://power.larc.nasa.gov/api/temporal/daily/point?parameters=T2M,RH...
 "type": "Feature",
 "geometry": {
  "type": "Point",
  "coordinates": [
   75.85,
   30.9,
   248.3
  ]
 },
 "properties": {
  "parameter": {
   "T2M": {
    "20241101": 22.1,
    "20241102": 21.6,
    "20241103": 21.2
   },
   "RH2M": {
    "20241101": 58.3,
    "20241102": 61.0,
    "20241103": 63.4
   },
   "WS2M": {
    "20241101": 1.4,
```

```
  ... (38 more lines in samples/nasa_power_daily.json)
```

FIELD	MEANING / HOW WE USE IT
`properties.parameter.<P>.<YYYYMMDD>`	value per day; −999 means missing (fill_value)
`parameters.<P>.units`	units; WS2M is m/s

4.14 OpenAQ — nearby stations

Call: `GET api.openaq.org/v3/locations?coordinates=lat,lon&radius=25000 (X-API-Key)`

Key: OpenAQ key **Freshness:** hourly **File:** `samples/openaq_locations.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_endpoint": "GET https://api.openaq.org/v3/locations?coordinates=30.9,75.85&radius=2500...
 "meta": {
  "name": "openaq-api",
  "website": "/",
  "page": 1,
  "limit": 5,
  "found": 2
 },
 "results": [
  {
   "id": 12345,
   "name": "Punjab Agricultural University, Ludhiana - PPCB",
   "locality": "Ludhiana",
   "timezone": "Asia/Kolkata",
   "country": {
    "id": 9,
    "code": "IN",
    "name": "India"
   },
   "isMobile": false,
   "isMonitor": true,
   "sensors": [
    {
     "id": 67890,
  ... (25 more lines in samples/openaq_locations.json)
```


FIELD	MEANING / HOW WE USE IT
`results[].sensors[]`	each sensor id + parameter (pm25 etc.)
`distance`	metres from your point
`datetimeLast`	last report — skip stale stations

4.15 OpenAQ — measurements

Call: `GET api.openaq.org/v3/sensors/{id}/measurements?datetime_from=…`

Key: OpenAQ key **Freshness:** hourly **File:** `samples/openaq_measurements.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_endpoint": "GET https://api.openaq.org/v3/sensors/67890/measurements?datetime_from=202...
 "meta": {
  "name": "openaq-api",
  "page": 1,
  "limit": 100,
  "found": 24
 },
 "results": [
  {
   "value": 182.0,
   "parameter": {
    "id": 2,
    "name": "pm25",
    "units": "ug/m3"
   },
   "period": {
    "label": "raw",
    "interval": "01:00:00",
    "datetimeFrom": {
     "utc": "2024-11-12T10:00:00Z",
     "local": "2024-11-12T15:30:00+05:30"
    },
    "datetimeTo": {
     "utc": "2024-11-12T11:00:00Z",
  ... (10 more lines in samples/openaq_measurements.json)
```


FIELD	MEANING / HOW WE USE IT
`results[].value`	concentration
`period.datetimeFrom.local`	local time of the hour

4.16 Overpass (OpenStreetMap)

Call: `POST overpass-api.de/api/interpreter data=[out:json];way[landuse=industrial](bbox);out center tags;`

Key: none **Freshness:** edited daily **File:** `samples/overpass_industrial.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_query": "[out:json][timeout:60];(way[\"landuse\"=\"industrial\"](30.7,75.6,31.1,76.1);...
 "version": 0.6,
 "generator": "Overpass API",
 "osm3s": {
  "timestamp_osm_base": "2026-10-20T05:59:00Z",
  "copyright": "The data included in this document is from www.openstreetmap.org. The dat...
 },
 "elements": [
  {
   "type": "way",
   "id": 123456789,
   "center": {
    "lat": 30.884,
    "lon": 75.921
   },
   "tags": {
    "landuse": "industrial",
    "name": "Focal Point Phase 8"
   }
  },
  {
   "type": "node",
```

```
  "id": 987654321,
  "lat": 30.951,
... (8 more lines in samples/overpass_industrial.json)
```

FIELD	MEANING / HOW WE USE IT
elements[].type	node / way / relation
center.lat/lon or lat/lon	location for distance decay F42
tags	landuse, man\{}_made=kiln, name ...

4.17 CelesTrak orbit elements

Call: `GET celestrak.org/NORAD/elements/gp.php?CATNR=43013&FORMAT=json`

Key: none **Freshness:** several times a day **File:** `samples/celestrak_gp_noaa20.json`


```
[
 {
  "_note": "Shape copied from the provider's official documentation. Values are illustrat...
  "_endpoint": "GET https://celestrak.org/NORAD/elements/gp.php?CATNR=43013&FORMAT=json",
  "OBJECT_NAME": "NOAA 20 (JPSS-1)",
  "OBJECT_ID": "2017-073A",
  "EPOCH": "2026-10-19T22:41:10.123456",
  "MEAN_MOTION": 14.1952,
  "ECCENTRICITY": 0.0001,
  "INCLINATION": 98.72,
  "RA_OF_ASC_NODE": 21.4,
  "ARG_OF_PERICENTER": 95.3,
  "MEAN_ANOMALY": 264.8,
  "EPHEMERIS_TYPE": 0,
  "CLASSIFICATION_TYPE": "U",
  "NORAD_CAT_ID": 43013,
  "ELEMENT_SET_NO": 999,
  "REV_AT_EPOCH": 45210,
  "BSTAR": 4e-05,
  "MEAN_MOTION_DOT": 4e-07,
  "MEAN_MOTION_DDOT": 0
 }
]
```


FIELD	MEANING / HOW WE USE IT
EPOCH	time the elements are valid for — older than 2 days = refresh
MEAN_MOTION, INCLINATION, …	orbit elements for satellite.js / skyfield
NORAD_CAT_ID	43013 NOAA-20; 37849 Suomi NPP; 54234 NOAA-21; 25994 Terra; 27424 Aqua

4.18 geoBoundaries (REAL response)

Call: `GET www.geoboundaries.org/api/current/gbOpen/IND/ADM2/`

Key: none **Freshness:** static **File:** `samples/geoboundaries_IND_ADM2_REAL.json`


```
{
 "_note": "REAL response fetched 20 Oct 2026 from https://www.geoboundaries.org/api/curre...
 "boundaryID": "IND-ADM2-76128533",
 "boundaryName": "India",
 "boundaryISO": "IND",
 "boundaryYearRepresented": "2021",
 "boundaryType": "ADM2",
 "boundaryCanonical": "District",
 "boundarySource": "Pathways Data Pvt. Ltd., lgdirectory.gov.in",
 "boundaryLicense": "Open Data Commons Open Database License 1.0",
 "sourceDataUpdateDate": "Thu Jan 19 07:31:04 2023",
 "buildDate": "Dec 12, 2023",
 "admUnitCount": "736",
 "meanAreaSqKM": "4449.7088250729",
 "gjDownloadURL": "https://github.com/wmgeolab/geoBoundaries/raw/9469f09/releaseData/gbOp...
 "simplifiedGeometryGeoJSON": "https://github.com/wmgeolab/geoBoundaries/raw/9469f09/rele...
}
```


FIELD	MEANING / HOW WE USE IT
`admUnitCount`	736 districts
`gjDownloadURL` / `simplifiedGeometryGeoJSON`	files to download once
`boundaryYearRepresented`	2021 — newer districts may be missing

4.19 LLM chat (OpenAI-compatible)

Call: `POST {base}/v1/chat/completions`

Key: provider key **Freshness:** - **File:** `samples/llm_chat_openai_compatible.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_request": {
  "model": "llama-3.3-70b-versatile",
  "messages": [
   {
    "role": "system",
    "content": "Write a 3-line brief. Use only numbers from the JSON."
   },
   {
    "role": "user",
    "content": "{\"district\":\"Ludhiana\",\"h\":14,\"lo\":12,\"hi\":16}"
   }
  ],
  "temperature": 0.2
 },
 "id": "chatcmpl-abc123",
 "object": "chat.completion",
 "created": 1760940000,
 "model": "llama-3.3-70b-versatile",
 "choices": [
  {
   "index": 0,
   "message": {
    "role": "assistant",
    "content": "Ludhiana recorded 14 fire cell-days (range 12-16) ..."
  ... (10 more lines in samples/llm_chat_openai_compatible.json)
```

FIELD	MEANING / HOW WE USE IT
`choices[0].message.content`	the text answer
`usage`	tokens used — log it to watch free limits
`model`	which model really answered (routers may switch)

4.20 Gemini generateContent

Call: `POST generativelanguage.googleapis.com/v1beta/models/{model}:generateContent`

Key: Gemini key **Freshness:** - **File:** `samples/gemini_generate_content.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_endpoint": "POST https://generativelanguage.googleapis.com/v1beta/models/{model}:gener...
 "candidates": [
  {
   "content": {
    "parts": [
     {
      "text": "Ludhiana recorded 14 fire cell-days ..."
     }
    ],
    "role": "model"
   },
   "finishReason": "STOP",
   "index": 0
  }
 ],
 "usageMetadata": {
  "promptTokenCount": 58,
  "candidatesTokenCount": 35,
  "totalTokenCount": 93
 },
 "modelVersion": "<flash model name shown in AI Studio>"
}
```


FIELD	MEANING / HOW WE USE IT
`candidates[0].content.parts[0].text`	the answer
`usageMetadata`	token counts

4.21 Groq Whisper transcription

Call: `POST api.groq.com/openai/v1/audio/transcriptions`

Key: Groq key **Freshness:** - **File:** `samples/groq_whisper_transcription.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "_endpoint": "POST https://api.groq.com/openai/v1/audio/transcriptions (multipart: file,...
 "task": "transcribe",
 "language": "hindi",
 "duration": 6.4,
 "text": "Ludhiana mein kal kitni aag lagi?",
 "segments": [
  {
   "id": 0,
   "start": 0.0,
   "end": 6.4,
   "text": "Ludhiana mein kal kitni aag lagi?",
   "avg_logprob": -0.21,
   "no_speech_prob": 0.01
  }
 ]
}
```

FIELD	MEANING / HOW WE USE IT
`text`	transcript
`language`	detected or forced (hi = Hindi)
`segments[]`	timestamps per sentence

4.22 Embeddings (OpenAI-compatible)

Call: `POST {base}/v1/embeddings`

Key: provider key **Freshness:** - **File:** `samples/embeddings_openai_compatible.json`


```
{
 "_note": "Shape copied from the provider's official documentation. Values are illustrati...
 "object": "list",
 "data": [
  {
   "object": "embedding",
   "index": 0,
   "embedding": [
    0.0123,
    -0.0456,
    0.0789,
    "... (384 to 3072 numbers depending on model)"
   ]
  }
 ],
 "model": "<embedding model>",
 "usage": {
  "prompt_tokens": 12,
  "total_tokens": 12
 }
}
```


FIELD	MEANING / HOW WE USE IT
`data[].embedding`	vector of numbers — store in pgvector / Qdrant
`usage`	tokens

4.23 Our own API (the contract)

Call: `GET /api/cells/1209_2558?month=2024-11`

Key: none (public read) **Freshness:** - **File:** `samples/firework_api_cell.json`


```
{
 "_note": "OUR OWN API response (the contract every page reads).",
 "data": {
  "cell": "1209_2558",
  "month": "2024-11",
  "fire_days_viirs_eq": 14,
  "lo": 12,
  "hi": 16,
  "clear_days": 22,
  "rate": 0.64,
  "unit": "VIIRS-equivalent fire cell-days",
  "flags": [
   "blind_spot"
  ]
 },
 "meta": {
  "data_source": {
   "fire_points": "FIRMS archive VNP14IMG + MCD14ML",
   "clear_days": "GEE MYD14A1",
   "burned_area": "GEE MCD64A1"
  },
```

```
  "fallback_used": false,
  "version": "calib-0.3",
  "generated_at": "2026-10-20T06:00:00Z",
  "demo": true
}
  ... (1 more lines in samples/firework_api_cell.json)
```

FIELD	MEANING / HOW WE USE IT
`data`	the numbers the page shows
`meta.data_source`	which source answered each input
`meta.fallback_used`	true when a backup was used
`meta.demo`	true → show DEMO chip

TIME

5 What changes with time and what does not

Cache what is static, refresh what moves, version what gets reprocessed.

Some data never changes once published, some changes every hour, and some is quietly replaced by a better version months later. Treating all of them the same either wastes your free API limits (calling static data again and again) or shows wrong numbers (keeping stale data). The table below sets a rule for each one. The cache time is how long the backend may reuse an answer before asking the source again.

DATA	UPDATES	CAN IT CHANGE LATER?	CACHE	REMEMBER
Fire points NRT	every 3 h	replaced by SP after ~2–3 months	1 h	store with version column
Fire points archive (SP)	monthly batches	new collection versions are rare but happen (C6.1, VIIRS 002)	never (refresh monthly)	keep file checksums
Daily fire masks	daily	stable	never after export	export monthly tables
Burned area	monthly, 2–3 months late	stable	never after export	mark recent months 'pending'
Land cover	years	new editions	forever	store edition year
OSM industry	daily edits	people add/remove sites	30 days	save snapshot date
Weather forecast	hourly	forecast changes	1 h	-
Weather archive	daily	final after ~5 days	forever after 5 days	-
Air quality	hourly	stations corrected later	1 h	flag stale stations
Orbit elements	several per day	drift	6–12 h	store EPOCH
Boundaries	rare	new districts	forever	store build date
LLM / AI answers	per request	models updated without notice	cache by prompt hash 24 h	log model name

Where the accurate data is

For history, the **standard-processing archive** (MODIS C6.1, VIIRS collection 2) is the accurate version; NRT is faster but provisional. For clear-sky counts, use the **daily L3 fire masks** rather than guessing clouds from other products. For burned area, **MCD64A1** is the standard global product; FireCCI is an independent check. For stations, prefer **OpenAQ reference-grade monitors** and show low-cost sensors separately.

BACKUPS

6 When a source fails

A drill for every input.

Free services fail in predictable ways: rate limits (HTTP 429), expired or wrong keys (401/403), maintenance (5xx), slow queues, and quiet changes in format. The resolver handles the first three automatically by moving to the next source and remembering the failure for ten minutes. The table below is the manual drill the team follows during the hackathon.

FAILURE	WHAT HAPPENS / WHAT TO DO
FIRMS API 429	wait for the 10-min window; use the stored archive; reduce bbox and days
FIRMS down	serve last stored data with a CACHED chip; LAADS files as backup
Earth Engine quota (restricted mode)	use exported tables (already in storage); Planetary Computer for new months
Overpass busy	mirror server; Geofabrik extract offline
Open-Meteo down	NASA POWER; then ERA5
OpenAQ key limit	Open-Meteo air quality (model, labelled); WAQI
CelesTrak down	Space-Track; keep yesterday's elements (valid ~2 days)
LLM provider limit	next provider in the router; template text if all fail (numbers never come from the LLM)
Format changed	schema check fails → resolver rejects → backup; open an issue

Ek source gira? Koi baat nahi — agla source khud aa jayega, aur page par likha hoga kaunsa.

6.1 Data kit files

The zip that comes with this PDF contains: `data_sources.json` (the registry), `fx_engine.py` (the resolver, tested offline with a failing primary source and a working backup), `fetch_samples.py` (saves real responses with your keys), `samples/` (25 response examples) and `README.md` (this chapter in short form).